

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize in the Rocq prover two operational semantics for PROLOG with cut and prove them equivalent. The first semantics is based on a stack of alternative computations to explore and is well suited for studying optimizations employed by standard PROLOG abstract machines; moreover, it closely models ELPI’s implementation. The second semantics is instead based on an and-or tree, in which the branches pruned by the *cut* operator are made explicit.

We use the tree-based semantics to reason about the determinacy analysis introduced in Elpi version 3.0, which statically identifies computations that produce, at run time, at most one result.

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1 Introduction

ELPI is a dialect of λ PROLOG (see [14, 15, 7, 12]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [13, 3, 4, 19, 8, 9]. Examples include the Hierarchy-Builder library-structuring tool [5] and Derive [17, 18, 11], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI gained a static analysis for determinacy [10] to help users tame backtracking. ROCQ users are familiar with functional programming but not necessarily with logic programming and uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checkers identifies predicates that behave like functions, i.e., predicates that commit to their first solution and leave no *choice points* (places where backtracking could resume).

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [10]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantic depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI’s implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [16] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations used by standard PROLOG abstract machines [20, 1], but it makes reasoning about the scope of *cut* difficult. To address that limitation we introduce a tree-based semantics in which the branches pruned by *cut* are explicit and we prove the two semantics equivalent. Using the tree-based semantics we then show that if every rule of a predicate passes the determinacy



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Inductive P := IP of nat. Inductive D := ID of nat. Inductive V := IV of nat.

Inductive Tm :=
  | Tm_P of P
  | Tm_D of D
  | Tm_V of V
  | Tm_App of Tm & Tm.

Record Unif := {
  unify : Tm → Tm → Σ → option Σ;
  matching : Tm → Tm → Σ → option Σ;
}.

```

■ Figure 1 Tm types

■ Figure 2 Unif type

44 analysis, the call to a deterministic predicate does not leave any choice points.

45 2 Common code: the language

put unif and proe Before going to the two semantics, we show the piece of data structure that are shared
 gram in variable: by the them. The smallest unit of code that we can use in the language is an atom. The
 hides from types atom inductive (see Type 1) is either a cut or a call. A call carries a term (see Figure 1). A
 term (Tm) is either a predicate, a datum, a variable or the binary application of two terms.

```

50 Inductive A := cut | call : Tm → A. (1)
51 Record R := mkR { head : Tm; premises : list A }. (2)
52 Record P := { rules : seq R; sig : sigT }. (3)
53 Definition Σ := {fmap V → Tm}. (4)
54 Definition bc : P →  $\mathcal{F}_V$  → Tm → Σ →  $\mathcal{F}_V$  * seq (Σ * seq A) := (5)

```

55 A rule (see Type 2) has a head of type term and a list of premises, that are a list of
 56 atoms. A program (see Type 3) is made by its rules and the signatures of predicates, **sigT**
 57 is a mapping from predicates to their signatures, signatures represent the definition of type
 58 from the simply typed lambda calculus, i.e. it is either a base type or an arrow for term
 59 application. We decorate arrows to know the mode (input or output) of the application
 60 argument.

61 A substitution (see Type 4) is a mapping from variables to terms. It is the output of a
 62 successful query. The empty substitution is noted ϵ .

63 The backchain function (bc, see Type 5) takes: a program P ; a set fv of variables ($\mathcal{F}_V$); a
 64 query q ; and the substitution σ . A backchain is performed when the current atom is a **call**.
 65 In this case, it start by recursively *dereferencing* the variables in σ , i.e. replacing all the
 66 variables in the query by their term mapping in σ .

67 TODO: free variables and program freshness

68 Then it takes the head of the term, which should be rigid, i.e. a predicate p , and look
 69 for the signature s of p in the program. Then, for each rule r in P , it unifies the head of r
 70 with q . Unification is possible thanks to the unificator U , whose type is shown in Figure 2.
 71 U explains how to unify terms up to standard unification (for output terms) or matching
 72 (for input terms). The function *unif* (resp. *matching*) unifies (resp. matches) the two terms
 73 under σ , it return an extension of σ if the two terms can be unified (resp. matched), and
 74 None otherwise. The result of a backchain is made by filtering the rules in P that can be used
 75 on the given query. In particular, for each rule filtered rules r , it returns a pair containing
 76 the extension of σ making the head of r and q unify and the list of premises of r .

```

f 1 2.   f 2 3.   r 0 0.   r 0 1.   r 0 2.
g X Z :- r X Y, f Y Z, !. % r1
g X Z :- f X Y, f Y Z.    % r2

```

Inductive tree :=
 | KO | OK | TA of A
 | Or of option tree & Σ & tree
 | And of tree & seq A & tree.

■ **Figure 3** Small ELPI program example

■ **Figure 4** Tree inductive

2.1 The cut operator

ELPI is a dialect of λ PROLOG with the hard cut operator and, in the language we have shown before, *cut* is an atom. The semantics of the cut operator adopted in the ELPI language corresponds to the *hard cut* operator of standard SWI-PROLOG. This operator has two primary purposes. First, it eliminates all alternatives that are created either simultaneously with, or after, the introduction of the cut into the execution state.

The ELPI program we will use as an example in the following sections is shown in Figure 3.

To illustrate the high-level description of *cut*, consider the program P shown in Figure 3 and the query $q = g \ 0 \ R$. Both rules for g can be used on the query q . Backchaining q in P from the empty (ϵ) substitution returns the list $[(\sigma_1, [r \ X \ Y, f \ Y \ Z, !]), (\sigma_2, [f \ X \ Y, f \ Y \ Z])]$ with σ_1 and σ_2 both equal to $\{X \mapsto 0, Z \mapsto R\}$. The two elements of the list are generated respectively from the rules r_1 and r_2 .

The next step of the interpretation corresponds to the backchaining of $r \ X \ Y$ from σ_1 . This returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [])]$ with $\sigma_3 := \sigma_1 + \{Y \mapsto 0\}$, $\sigma_4 := \sigma_1 + \{Y \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{Y \mapsto 2\}$. The three lists of premises are empty since all rules for r have no premises.

Backchaining $f \ Y \ Z$ in σ_3 gives no solution, therefore, by backtracking, we backchain $f \ Y \ Z$ in σ_4 . This gives a solution, namely $\sigma_6 := \sigma_4 + \{R \mapsto 2\}$.

We now reach the cut operator and can see its double side behavior. In the current execution state, we have still two unexplored alternatives: we can backtrack on the third solution for r that assign Y to 2 and on the rules r_2 . We can note, however, the first alternative have been generated after the cut introduction, and the second alternative have been generated at the same moment of the cut. Therefore both are cut away leaving no choice point. The final complete solution is: $\{X \mapsto 0, Z \mapsto R, Y \mapsto 1, R \mapsto 2\}$, where X , Y and Z can be ignored since they are the result internal unification.

3 Semantics intro

We propose two operational semantics for a logic program with cut. The two semantics are based on different syntaxes, the first syntax (called *tree*) exploits a tree-like structure and is ideal both to have a graphical view of its evolution while the tree is being interpreted and to prove lemmas over it. The second syntax, called *elpi*, is the ELPI's syntax and has the advantage of reducing the computational cost of cutting and backtracking alternatives by using shared pointers. We aim to prove the equivalence of the two semantics together with some interesting lemmas of the cut behavior.

4 Tree semantics

The tree state The tree inductive, shown in Figure 4. We distinguish 5 main cases: KO , OK , and are special meta-symbols representing, respectively, the basic failed and a successful

se metti $r1 = g \ A$
 $B :- f \ A \ B$. allora
 $g \ e \ f$ sono fun-
 zioni, e puoi spie-
 gare anche l'idea
 del detcheck qui

```

Fixpoint get_end s A :  $\Sigma$  * tree :=
  match A with
  | TA _ | KO | OK  $\Rightarrow$  (s, A)
  | Or None s1 B  $\Rightarrow$  get_end s1 B
  | Or (Some A) _ _  $\Rightarrow$  get_end s A
  | And A _ B  $\Rightarrow$ 
    let (s', pA) := get_end s A in
    if pA == OK then get_end s' B
    else (s', pA)
  end.

Definition get_subst s A := (get_end s A).1.
Definition path_end A := (get_end  $\epsilon$  A).2.
Definition success A := path_end A == OK.
Definition failed A := path_end A == KO.
Definition path_atom A :=
  if path_end A is TA _ then true
  else false.

```

(a) Definition of *get_end*(b) Functions from *get_end*

■ **Figure 5** *get_end* and derives functions

113 trees. These symbols are considered meta because they are internal intermediate symbols
 114 used to give structure to the tree.

115 The *TA* constructor (acronym for tree-atom) is the constructor of atoms in the tree. They
 116 are either the cut operator or a call, in these cases, the interpreter should evaluate the atom
 117 by either cutting the subtrees in the scope of the cut or backchaining the call in the program.

118 The two recursive cases of a tree are the *Or* and *And* nodes. The *Or* node $A \vee B_\sigma$ denotes
 119 a disjunction between two trees *A* and *B*. The first branch is optional, if absent it represents
 120 a dead tree, i.e. a tree that has been entirely explored. The second branch is annotated
 121 with a suspended substitution σ so that, upon backtracking to *B*, σ is used as the initial
 122 substitution for the execution of *B*.

123 The *And* node $A \wedge_{B_0} B$ represents a conjunction of two trees *A* and *B*. We call B_0 the
 124 reset point for *B*; it is used to restore the state of *B* to its initial form if a backtracking
 125 operation occurs on *A*. An example of this behavior is shown in Section 4.2

126 A graphical representation of a tree is shown in Figure 6a. To make the graph more
 127 compact, the *And* and *Or* nodes are n-ary rather than binary, and the children of these
 128 nodes associate to the right. The *KO* node acts as the neutral elements in the *Or* list, while
 129 *OK* is the neutral element of the *And* list.

130 *The tree interpreter* The interpretation of a tree is performed by two main routines: *step*
 131 and *next_alt* that traverse the tree depth-first, left-to-right. Then, then *runT* inductive
 132 makes the transitive closure of *step* and *next_alt*: it iterates the calls to its these
 133 auxiliary functions as much as needed. In Types 7–9 we give the types contrats of these
 134 symbols where $\mathcal{F}_v$ is a set of variable names.

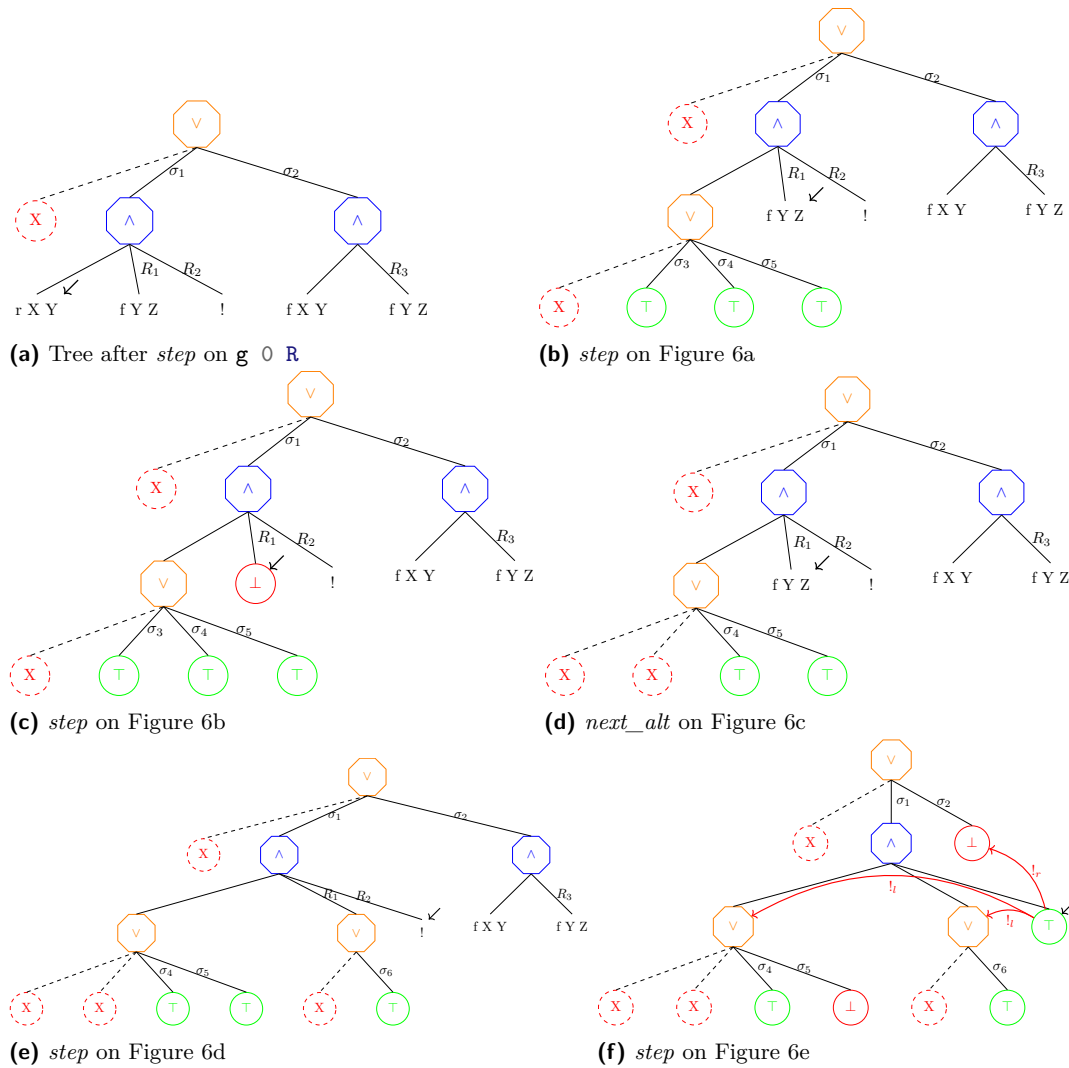
135 **Inductive** tag := Expanded | CutBrothers | Failed | Success. (6)

136 **Definition** step : $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F}_v * \text{tag} * \text{tree}) :=$ (7)

137 **Definition** next_alt : $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=$ (8)

138 **Inductive** runT (p : $\mathbb{P}$) : $\mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow$
 $\Sigma \rightarrow \text{option tree} \rightarrow \text{Prop} :=$ (9)

139 The tree interpreter, as in prolog, explores the tree in DFS strategy, to discover the
 140 substitution and the leaf of the tree that should be interpreted. These two pieces of
 141 information can be retrieved for a substitution *s* and a tree *A* thanks to the *get_end* routine,
 142 shown in Figure 5a If the tree is already a leaf, *get_end* returns its inputs. Otherwise, if the
 143 tree is a disjunction, the path continues on the left subtree, if it exists, otherwise *get_end*
 144 contues on the ths using the substitution stored in the *Or* node. In the case of a conjunction,



■ **Figure 6** Some tree representations¹

145 if *get_end* of the lhs returns the tree *OK*, then the *get_end* continues in the rhs, otherwise
 146 we return the result of the lhs. We derive some useful definition from *get_end*. They are
 147 listed in Figure 5b.

148 We have pointed by a black arrow the *path_end* of all the trees in Figure 6.

149 4.1 The *step* procedure

150 The *step* procedure takes as input a program, a set of variables, a substitution, and a tree,
 151 and returns an updated set of variables, a *tag*, and an updated tree.

152 The set of variables v given in input to *step* are used and updated when a backchaining
 153 operation takes place. A call to *step* produces a good result if v contains all the variables

¹ The substitution $\sigma_1 \dots \sigma_6$ are from Section 2.1, R_1, R_2 and R_3 are the reset points and correspond respectively to the list of atoms $[f \ Y \ Z, !], [!]$ and $[f \ Y \ Z]$

in the tree. This guarantees that any call to *bc* returns a list of atoms whose variables are disjoint from the variables of the current working tree.

The *tag* (see Type 6) indicates the kind of an internal tree step: **CutBrothers** denotes the interpretation of a superficial cut, i.e., a cut whose parent nodes are all *And*-nodes. **Expanded** denotes the interpretation of non-superficial cuts or predicate calls. **Failure** and **Success** are returned for, respectively, *failed* and *success* trees.

The step procedure is intended to evaluate atoms, that is, it leaves unchanged the tree iff its *path_end* is not an atom, i.e. if it is a success- or a failed-tree.

Lemma *succ_step_iff* $u\ p\ v\ s\ A$: $\text{success } A \leftrightarrow \text{step } u\ p\ v\ s\ A = (v, \text{Success}, A)$. (1)

Lemma *fail_step_iff* $u\ p\ v\ s\ A$: $\text{failed } A \leftrightarrow \text{step } u\ p\ v\ s\ A = (v, \text{Failed}, A)$. (2)

The interpretation of a call to a term *c* starts by calling the *bc* function on *c*. The result of the backchain is a list *l*. If *l* is empty then the current call is replaced by the *KO* node, otherwise it is replaced by a right skewed tree made by *Or* inner-nodes and each leave correspond to a list of atom *e* in *l*. If *e* is empty then it is mapped to the *OK* tree, otherwise it is mapped to a right-skewed tree made by *And* inner-nodes.

In Figure 6a, we provide a clear example of a call to *step* on the query $q\ 0\ R$ on the program *P* in Figure 3. Let *c*₀, *c*₁ and *c*₂ be the children of the root. *c*₀, by construction is the *None* tree, while *c*₁ and *c*₂ correspond respectively to the premises of rules *r*₁ and *r*₂ in *P*. We need the first *None* subtree so that we can attach a substitution to *c*₁. In particular, we note that the *c*₁ (resp. *c*₂) are related to the substitution σ_1 (resp. σ_2). The value of σ_1 and σ_2 are the same as in Section 2.1. The reset points in the tree are marked with the *R* symbol. $R_1 := [f\ Y\ Z, !]$, $R_2 := [!]$ and $R_3 := f\ Y\ Z$, they are essentially the list of atoms that allowed to generate the current subtree. Other example of call to the *step* procedure triggering a backchain are Figures 6b, 6c, and 6e.

The interpretation of a cut has three main impacts: at first the cut is replaced by the *OK* node, then some special subtrees, in the scope of the *Cut*, are cut away: in particular we need to soft-kill the left-siblings of the *Cut* and hard-kill the right-uncles of the *Cut*.

► **Definition 1** (Left-siblings (resp. right-sibling) and Right-uncles). Given a node *A*, the left-siblings (resp. right-sibling) of *A* are the list of subtrees sharing the same parent of *A* and that appear on its left (resp. right). The right-uncles of *A* are the list of right-sibling of the father of *A*.

► **Definition 2** (Hard-kill, $!_r$ and Soft-kill, $!_l$). Given a tree *t*, hard-kill replaces the given subtree with the *KO* node. If *t* is a success-tree, then soft-kill replaces with *KO* all subtrees that are not part of the path in *t* leading to the *OK* node.

An example of the impact of the cut is show in Figure 6f. In Figure 6e, we see that the *path_end* is a cut. The cut is replaced by the *OK* node and it need to cut-away some alternatives in the tree. The red arrows indicate the scope of the cut, they are labeled with $!_r$ and $!_l$ to explain which kind of cut has been performed on each pointed subtree. We note that, after the evaluation of a cut, the path leading to the cut node is not affected by $!_l$, the success under the substitution σ_4 is kept, while the success under the substitution σ_5 is replaced by the *KO* node. If we refer to the paragraph in Section 2.1, we are discarding the third rule of the *r* predicate.

4.2 The next_alt procedure

The *step* procedure alone is not sufficient to reproduce entirely the behavior of the full expected prolog interpreter. As proved in Lemma 1 and Lemma 2, *step* does not make any

progression on the tree being interpreted. To solve this problem, we need a routine capable of backtracking in case of failures. This routine return the backtracked tree if it exists, *None* otherwise.

In Figure 6c, we are facing a failure: the interpreter has evaluated the atom $\mathbf{f} \ X \ Y$ under the substitution σ_3 which assigns Y to 0, but, in Figure 3, we have no rule satisfying this query. The role of *next_alt*, in this example, is to kill the path leading to this failure, while resetting the failure to its original shape. We see how the reset point plays a major role in this case. At first we are killing the *OK* node under the substitution σ_3 , since this tree have been entirely explored and led to no solution, then we replace the *KO* node with the information stored in R_1 . That is, thanks to R_1 , we are restoring the $\mathbf{f} \ Y \ Z$ call, so that we are able to evaluate this node under the substitution σ_4 . More concretely, we have uncessefully tried rule $\mathbf{r} \ 0 \ 0$, therefore, we need to continue by trying rule $\mathbf{r} \ 0 \ 1$.

Furthermore, *next_alt* accomplishes to another important task. Its behavior changes depending to the value of the boolean it receives in entry: *next_alt false A* aims to recursively cancel all the failures (if any) in A , whereas, *next_alt true A* aims to cancel the first success in A . This second behavior is helpful since, we need to progress the tree in case of success. For example, consider the tree in Figure 6f, it contains a success, mapping namely R to 2. The interpreter should return this success, but, due to the “non-deterministic” behavior of the prolog system, we also need to return all the other solution that could be available by evaluating the rest of the tree. After the replacement of the *path_end* in Figure 6f with *KO*, we remain with a tree having all failing path. The *next_alt* procedure keeps removing them until discovering that the tree have no worth-to explore paths, therefore it will return *None*.

Some interesting property of *next_alt* are shown below and allow to see how *next_alt* complements *step* wrt failed, success and *path_atom* trees.

Lemma *path_atom_next_alt_id* $b \ A$: $\text{path_atom } A \rightarrow \text{next_alt } b \ A = \text{Some } A$. (3)

Lemma *next_alt_failedF* $b \ A \ A'$: $\text{next_alt } b \ A = \text{Some } A' \rightarrow \text{failed } A' = \text{false}$. (4)

4.3 The *runT* inductive

The inductive *runT* is modeled as a function: it takes as input a program, a set of variables, an initial substitution σ_0 , and a tree t_0 , and returns a substitution σ_1 together with an optional updated tree t_1 . The substitution σ_1 represents the most-general unificator that makes the execution of the tree t_0 succeed starting from the initial substitution σ_0 , σ_1 is an extension of σ_0 . The tree t_1 is the updated tree containing the alternatives that have not yet been explored. If the tree contains no solution, then *None* is returned.

The procedure *runT* is based on three main derivation rules, shown in Figure 7. If the *path_end* of the tree t is a success, the substitution is retrived in t from σ_0 and the input tree is cleaned of its successful path. If the *path_end* of the tree is an atom, then *step* is invoked to evaluate this atom, and *runT* is recursively called on the new tree. Finally, if the *path_end* of the tree is a failure, *next_alt* is called to clear the failed path; if the resulting cleaned tree exists, *runT* is recursively called on it.

In our implementation, *runT* gives the guarantee that the program interpretation terminates: it always return a substitution. A diverging program will never provide a solution. However, we can easily add a rule to treat this scenarion, but studying diverging programs in our settings is not particularly worthy.

$$\begin{array}{c}
\frac{\text{success } A \quad \text{get_subst } s_0 \ A = s_1 \quad \text{next_alt true } A = B}{\text{runT } v_0 \ s_0 \ A \ s_1 \ B} \text{ RUN_DONE} \\
\\
\frac{\text{path_atom } A \quad \text{step u p } v_0 \ s_0 \ A = (v_1, \text{st}, B) \quad \text{runT } v_1 \ s_0 \ B \ s_1 \ C}{\text{runT } v_0 \ s_0 \ A \ s_1 \ C} \text{ RUN_STEP} \\
\\
\frac{\text{failed } A \quad \text{next_alt false } A = \text{Some } B \quad \text{runT } v_0 \ s_0 \ B \ s_1 \ C}{\text{runT } v_0 \ s_0 \ A \ s_1 \ C} \text{ RUN_FAIL}
\end{array}$$

■ **Figure 7** Rule system for *runT*

242 4.3.1 Properities of *runT*

243 5 Elpi semantics

244 We now introduce the ELPI semantics. The interpreter is close to that of the ELPI language
 245 and provides an operational semantics that is similar to the one proposed by Pusch in [16],
 246 which is in turn closely related to the semantics given by Debray and Mishra in [6, Section 4.3].
 247 Pusch mechanized this semantics in Isabelle/HOL, together with several optimizations that
 248 are also present in the Warren Abstract Machine [20, 1].

249 We represent a state of the ELPI language thanks to the two mutual inductives shown
 250 below.

```

Inductive alts := no_alt | more_alt of (Σ * goals) & alts
with goals := no_goals | more_goals of (A * alts) & goals .

```

251 An ELPI state is an enhanced two-dimensional list. The outermost list represents a list
 252 of alternatives in disjunction, each accompanied by the substitution that should be used
 253 for their interpretation. The innermost list is a list of atoms, that is, the list of goals in
 254 conjunction. These goals are decorated with a pointer to an alternative list, which is used to
 255 keep track of where to jump when a cut is interpreted. We call these special alternatives the
 256 *cut-to* alternatives.

257 The idea of the ELPI interpreter is to receive a list of alternatives. The first alternative
 258 consists of a list of goals. Four cases must be taken into account; they are shown in
 259 Figure 8. In order to simplify goal retrieval, we split the head of the alternatives from the
 260 tail, so that it can be immediately matched in the inductive definition. Note that an empty
 261 list of alternatives represents, by definition, a failing elpi state. If the goal list is empty
 262 (STOPE), then we have, by definition, a success, and the input solution together with the
 263 list of alternatives is returned. If the goal list starts with a cut (CUTE), then the current
 264 alternatives are erased in favour of the cut alternatives, and a recursive call is made on the
 265 remaining goal list.

266 Finally, we must consider the case in which the goal list starts with a call. The call
 267 can either fail (FAILE) or succeed (CALLE). We distinguish the two cases by looking if the
 268 backchaining operation returns zero or more rules. We have wrapped this task in the *stepE*
 269 procedure, which also updates the goal and cut-alternative list. The fail case, is relatively
 270 easy: the first goal does not succeed, we need to take the head of the alternatives, and make
 271 it the new list of goals to be explored.

272 The case in which backchaining produces a non empty list, the *save_alts* routine is in
 273 cahrg of: taking the list of premises and add to each atom the the list of alternatives *a* as
 274 their new cut-alternatives, then it append the list of goals *gl* to each of these new lists.

Definition $\text{stepE } v_0 \ t \ s \ a \ \text{gl} :=$
 $\text{let } (v_1, \text{rs}) := \text{bc } p \ v_0 \ t \ s \ \text{in}$
 $\text{let } \text{rs_ca} := \text{save_alts } a \ \text{gl} \ (\text{r2a } \text{rs}) \ \text{in}$
 $(v_1, \text{rs_ca}).$

Inductive $\text{runE} : \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{goals} \rightarrow \text{alts} \rightarrow \Sigma \rightarrow \text{alts} \rightarrow \text{Prop} :=$ (10)

$$\frac{}{\text{runE } v_0 \ s_0 \ [::] \ a \ s_0 \ a} \text{STOPE}$$

$$\frac{\text{runE } v_0 \ s_0 \ \text{gl} \ \text{ca} \ s_1 \ r}{\text{runE } v_0 \ s_0 \ [:: (\text{cut}, \text{ca}) \ \& \ \text{gl}] \ a \ s_1 \ r} \text{CUTE}$$

$$\frac{\text{stepE } v_0 \ t \ s_0 \ \text{al} \ \text{gl} = (v_1, [:: b \ \& \ \text{bs}]) \quad \text{runE } v_1 \ b.1 \ b.2 \ (\text{bs}++\text{al}) \ s_1 \ r}{\text{runE } v_0 \ s_0 \ [:: (\text{call } t, \text{ca}) \ \& \ \text{gl}] \ \text{al} \ s_1 \ r} \text{CALLE}$$

$$\frac{\text{stepE } v_0 \ t \ s_0 \ \text{al} \ \text{gl} = (v_1, [::]) \quad \text{runE } v_1 \ s_1 \ a \ \text{al} \ s_2 \ r}{\text{runE } v_0 \ s_0 \ [:: (\text{call } t, \text{ca}) \ \& \ \text{gl}] \ [:: (s_1, a) \ \& \ \text{al}] \ s_2 \ r} \text{FAILE}$$

■ **Figure 8** Rule system for runE

6 Semantic equivalence

The equivalence between the two semantics is possible under two conditions: we need to work with “valid tree”, i.e. all of those trees that can be generated from a call. Secondly, we need to translate trees into elpi states. In the next to subsection we propose the two functions *valid_state* and *t2l*.

6.1 Valid trees

The inductive tree allows one to generate a large number of trees, some of which are not valid, in the sense that they cannot be produced starting from a given query. The class of valid trees is characterized by the function shown in Figure 9a.

While all base cases of tree are considered valid, we need to analyze carefully the cases for the *Or* and *And* constructors.

For the *Or* constructor, we distinguish two cases depending on whether the left-hand side (lhs) exists. If it does not exist, then the right-hand side (rhs) must be a valid tree. Otherwise, the lhs must itself be a valid tree, and the rhs is either the *KO* tree, since it may have been removed by the evaluation of a superficial cut in the lhs, or it has not yet been explored. In the latter case, it is a **base_or** tree, namely the right-skewed tree formed by a disjunction of conjunctions that is generated after a successful backchain to a call.

For the *And* constructor, the lhs is required to be a valid tree. The shape of the rhs depends on whether the lhs is a success. If the lhs is not successful, then the rhs has never been explored: the procedures *step* and *next_alt* modify the rhs only when the lhs succeeds. In this case, the lhs must be the right-skewed tree containing conjunctions of premises atoms, the reset point B_0 ensure what shape the rhs should have. If the lhs is a success tree, then the rhs can be modified by *step* and *next_alt*, therefore it must be a valid tree.

```

Fixpoint t2l A s0 bt :=
  match A with
  | OK           ⇒ [:: (s0, [::])]
  | KO           ⇒ [::]
  | TA a         ⇒ [:: (s0, [:: (a, [::]) ])]
  | Or None s1 B ⇒ add_ca_deep bt (t2l B s1 [::])
  | Or (Some A) s1 B ⇒
    let lB := t2l B s1 [::] in
    let lA := t2l A s0 lB in
    add_ca_deep bt (lA ++ lB)
  | And A B0 B ⇒
    let lA := t2l A s0 bt in
    let lB0 := a2g B0 in
    let lA := add_deep bt lB0 lA in
    if lA is [:: (s0, gs) & al] then
      let al := map (catr lB0) al in
      let lB := t2l B s0 (al ++ bt) in
      map (catl gs) lB ++ al
    else [::]
  end.
end.

Fixpoint valid_tree A :=
  match A with
  | TA _ | OK | KO ⇒ true
  | Or None _ B ⇒ valid_tree B
  | Or (Some A) _ B ⇒ valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A B0 B ⇒ valid_tree A &&
    if success A then valid_tree B
    else B == big_and B0
  end.

```

(a) Valid tree

(b) Tree to list

■ Figure 9 Valid tree and Tree to list

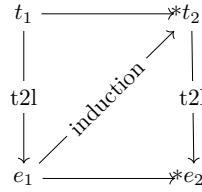
298 6.2 From trees to lists

299 The translation of a tree to a list is shown in Figure 9b. It takes the tree to be translated, a
 300 substitution (called s), a list of alternatives, called bt . The substitution s tells what is the
 301 subsitution of the alternative we are building and is updated when going to the rhs of the
 302 *Or* constructor. The bt list represents the future alternatives list that have the same depth
 303 of the current tree root. They are useful to know if they should be added to the current goal
 304 as cut alternatives.

305 An *OK* node represent a success in a tree, then the translation of this tree returns a
 306 singleton list with the cuple $(s, [::])$: there is one alternative with 0 goals, i.e. a success in
 307 elpi by STOPE. The *KO* node represent a failure, and, therefore 0 alternatives. An atom is
 308 translated into a singleton with the atom as first goal and the empty cut-alternative list: we
 309 are not adding bt since they are the alternatives for the right-sibling, not the right-uncles,
 310 this means that if we have a is a cut, then it erases the alternatives bt .

311 In a disjunction, we are scendendo di un livello the distance from our backtracking list.
 312 This means that bt becomes the list of right-uncles of the two branches of the *Or* constructor.
 313 The compilation of the rhs is done independently of if the lhs exists: we transform the rhs
 314 into an elpi state and passing the empty list of alternatives, since the rhs as no right-siblings.
 315 Then, thanks to `add_ca_deep`, we recursively add bt to every cut-alternative appearing in
 316 the translated goals. If the lhs exists, we translate it and pass the translation of the rhs as
 317 the list of right-siblings. Then we prepend the translated lhs to the translated rhs and add
 318 bt with `add_ca_deep`.

319 We compile the *And* case by translating its lhs to the list lA and reset point to the list
 320 $lB0$. Then, thanks to `add_deep`, we recursively append $lB0$ to the alternatives born in lA ,
 321 i.e. we leave unchanged the pointers to bt , if any, in lA . We finally need to treat cases
 322 whenever the list lA is empty or not. In the former case the empty list is returned, otherwise,



■ **Figure 10** Induction scheme for Lemma 6

we have a list $[(s_0, gs) \& al]$. By the semantics of the *And* in the *runT* interpreter, the rhs of the *And* represent the sequent of goals to be executed after the first alternative in the lhs, it corresponds to the list of goals *gs*. The reset point *B0* is to be appended to the next alternatives in the lhs, that is the queue of *lA*, that is *al*.

alternative in tree? rephrase

We note therefore that the compilation of *lB* is done by passing the alternatives *al* + *bt* since the alternatives in *al* are to be...

6.3 Equivalence theorems

We have now all the ingredients that ...

```

Lemma tree_to_elpi:  $\forall$  u p  $s_0$  t  $s_2$  t',
  let  $v_0 := \text{vars\_tree } t \setminus \text{vars\_sigma } s_0$  in
  valid\_tree t  $\rightarrow$ 
  runT u p  $v_0$   $s_0$  t  $s_2$  t'  $\rightarrow$ 
   $\exists$   $s_1$  g a,
    let  $na := \text{t2l } (\text{odflt } K0 \text{ t}') \ s_0 \ [::]$  in
     $\text{t2l } t \ s_0 \ [::] = (s_1, g) :: a \wedge$ 
    runE u p  $v_0$   $s_1$  g a  $s_2$  na.
  
```

```

Lemma elpi_to_tree:  $\forall$  u p  $v_0$   $s_1$  g  $s_2$  a na,
  runE u p  $v_0$   $s_1$  g a  $s_2$  na  $\rightarrow$ 
   $\forall$   $s_0$  t, valid\_tree t  $\rightarrow$   $\text{t2l } t \ s_0 \ [::] = (s_1, g) :: a \rightarrow$ 
   $\exists$  t', runT u p  $v_0$   $s_0$  t  $s_2$  t'  $\wedge$   $\text{t2l } (\text{odflt } K0 \text{ t}') \ s_0 \ [::] = na$ .
  
```

The proof of Lemma 6 is based on the idea explained in [2, Section 3.3]. An ideal statement for this lemma would be: given a function *l2t* transforming an elpi state to a tree, we would have that the execution of an elpi state *e* is the same as executing *runT* on the tree resulting from *l2t(e)*. However, it is difficult to retrieve the structure of an elpi state and create a tree from it. This is because, in an elpi state, we have no clear information about the scope of an atom inside the list and, therefore, no evident clue about where this atom should be placed in the tree.

Our theorem states that, starting from a valid tree *t* which translates to a list of alternatives $(\sigma_1, g) :: a$. If we run in elpi the list of alternatives, then the execution of the tree *t* returns the same result as the execution in elpi. The proof is performed by induction on the derivations of the elpi execution. We have 4 derivations.

We have 4 cases to analyse:

7 Case study: determinacy analysis

we mechanize the first order part of xxx.

snippet det, main thm, invariant det tree (valid tree prev section?)

proof induction on exec, step/next alt preserving invariant proved by induction on the tree.

with list semantics cut and next alt requires to express a link between the ca or next alts and the current goal, which is non trivial without an intermediate data structure like the tree

8 Related work

prolog semantics, King lost
yves for the proof technique

9 Conclusion

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